
Metadata as Epistemic Devices: Towards Weak Semantic Networks in Digital Archives

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Abstract

This paper asks whether metadata can function as epistemic devices - activating conceptual relations between heterogeneous digital sources and enabling any user, not only the specialist, to navigate an archive through historical reasoning. Until recently, the historiographical object was almost exclusively a book: primary sources appeared, when fortunate, as footnotes. Digital archives changed this asymmetry, yet accessibility is not interpretability. The answer proposed here is a weak semantic network: a lightweight, replicable model generating interpretive connectivity among sources without formal ontologies.

The case study is the We-Z digital archive on the history and memory of Vigne Nuove, a social housing district on the northern outskirts of Rome. Still in the implementation phase, the archive holds 81 digital objects: oral history video interviews (OH), documents from institutional archives including the Archivio Storico Capitolino and the Fondo Passarelli at the MAXXI (AS), and crowdsourced materials from residents (CS). The heterogeneity of these sources - born-digital, digitised, and community-produced - is the condition the model is designed to address.

Built on the OAIS logical model (Lavoie, 2004) and implemented in Omeka S using Dublin Core Metadata Initiative standard, the archive introduces conceptual metadata added to - not replacing - standard archival description. These are interpretive terms extracted through hermeneutic source analysis, drawing on Koselleck's (1985) conceptual history, assigned collaboratively by curators and, where possible, members of the reference community. Conceptual metadata function as nodes around which digital objects cluster through shared interpretive categories. Rather than computed through graph algorithms, the network emerges from iterative hermeneutic cycles: interpretation generates metadata, metadata reorganise retrieval, retrieval feeds back into interpretation. AI acts as a dialogic partner - what Yang et al. (2025) call metadata intelligence.

The central finding is that conceptual metadata produce bidirectional semantic movement: a concept illuminates documents, and those documents redefine the concept. In the We-Z archive, "trasformazione" returns a cluster connecting oral history interviews, archival planning documents, and crowdsourced photographs - sources sharing no formal ontological relation but interpretively connected through a term extracted from their contents. The same dynamic emerged at the 2022 Digital Humanities Summer School, where documents from different conflicts tagged with "cease-fire" - a conceptual metadata term identified by cross-referencing hermeneutic analysis with computational corpus analysis - formed a cluster clarifying the term's shifting legal and political meaning. The network is weak because it

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operates below the threshold of Semantic Web ontologies (OWL, SPARQL, RIC-O): the machine does not interpret queries, but the metadata structure activates meaningful navigation for human users. The archive is performative: users face a bare search bar and must formulate an interpretive act before the system responds. A conceptual lexicon under development will lower this threshold while preserving this demand (Fig. 1, Fig. 2).

The broader contribution is a replicable model for community digital archives in Public Digital History. Positioned between classical archival description and Semantic Web infrastructure, it requires no ontological expertise, uses open-source tools and an internationally recognised standard, already adopted by students in independent thesis projects. By making the interpretive path from category to source transparent and navigable, the archive renders historical method - not only historical content - publicly accessible.

Bibliography

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Figure Captions

Fig. 1 - The hermeneutic recursion model. From human and AI interpretation, conceptual metadata are produced and integrated into the OAIS-DCMI metadata layer of the digital object. The retrieval system feeds into the conceptual lexicon, which in turn enables new queries and reactivates the interpretive cycle. Credit: Manfredi Scanagatta.

Fig. 2 - Conceptual interoperability in the We-Z archive. The conceptual metadata term "nostalgia" functions as a shared node connecting Digital Objects of type OH (oral history), AS (archival sources), and CS (crowdsourced materials). Each object retains its standard OAIS-DCMI descriptive layer; the conceptual metadata operates as an additional relational edge. Credit: Manfredi Scanagatta.

Fig. 3 - The We-Z archive retrieval interface: a search for "trasformazione" returns 14 results across all three source types - oral history video interviews (OH), institutional archive documents (AS), and crowdsourced materials (CS) - interpretively connected through a shared conceptual metadata term. Credit: Manfredi Scanagatta / Progetto We-Z.

Keywords: Conceptual Metadata, Semantic Interoperability, Hermeneutic Recursion, Digital Archives, Artificial Intelligence, Public Digital History, Community Archives

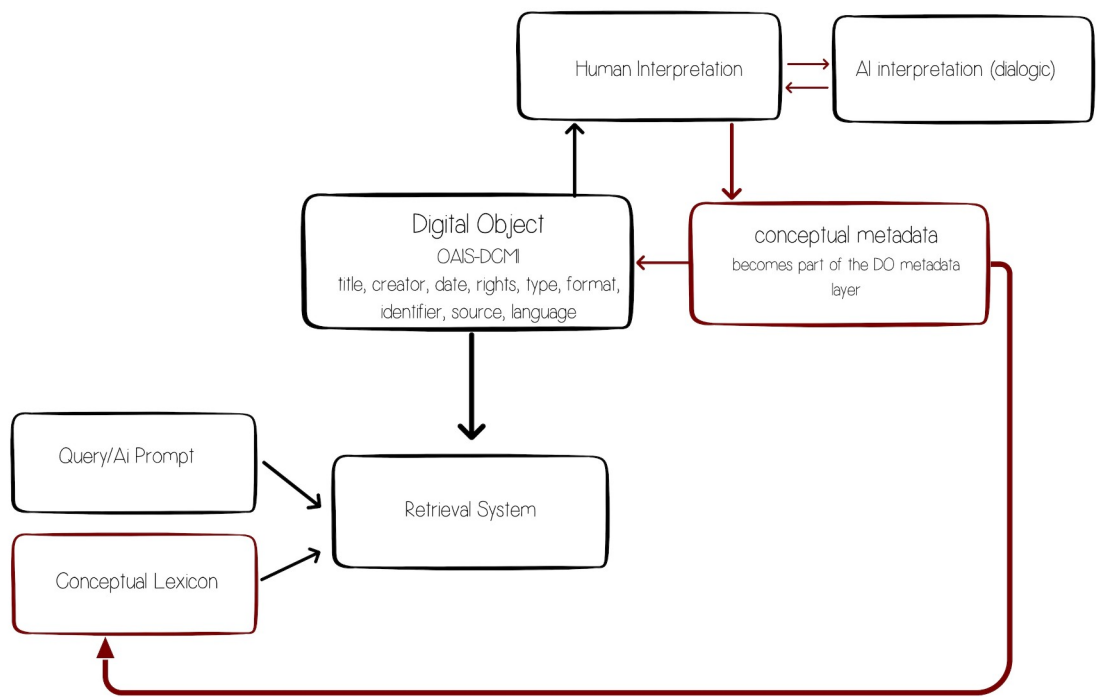


Figure 1. The hermeneutic recursion model. From human and AI interpretation, conceptual metadata are produced and integrated into the OAIS-DCMI metadata layer of the digital object. The retrieval system feeds into the conceptual lexicon, which in turn enables new queries and reactivates the interpretive cycle. Credit: Manfredi Scanagatta.

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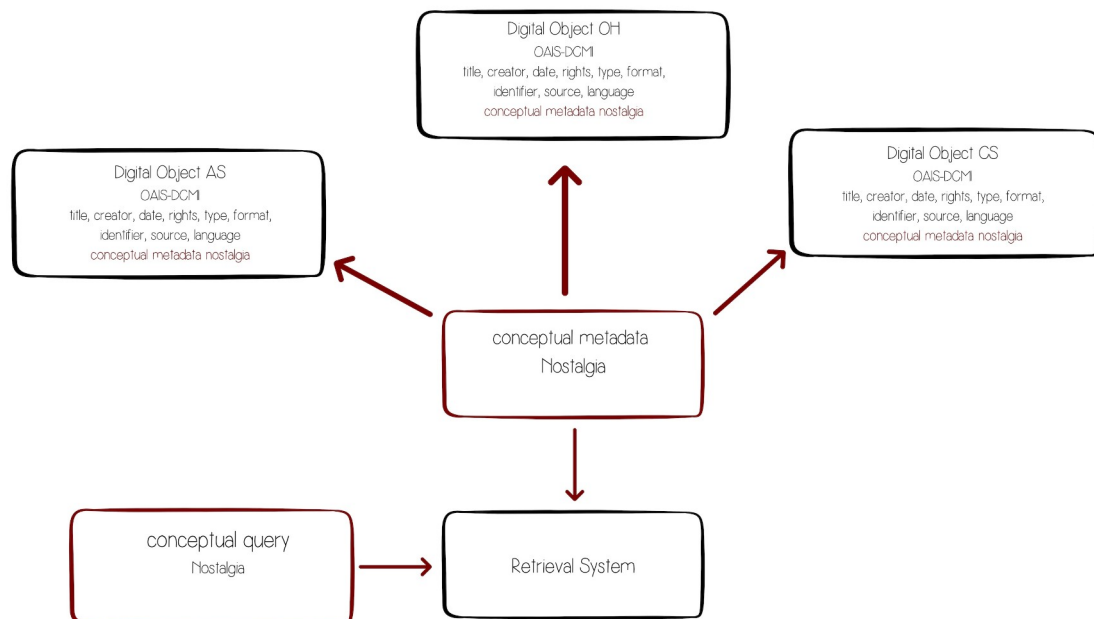


Figure 2. Conceptual interoperability in the We-Z archive. The conceptual metadata term "nostalgia" functions as a shared node connecting Digital Objects of type OH (oral history), AS (archival sources), and CS (crowdsourced materials). Each object retains its standard OAIS-DCMI descriptive layer; the conceptual metadata operates as an additional relational edge. Credit: Manfredi Scanagatta.

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Ricerca

Ricerca

trasformazione

CERCA

14 risultati per "trasformazione"

- 
[Video Intervista Dino Piras](#)
Lotta per la casa
Intervista a Dino Piras, attivista della lotta per la casa a Roma, che racconta la sua esperienza di militanza e la sua visione della città.
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Trascrizione dell'intervista a Dino Piras, attivista della lotta per la casa a Roma, che racconta la sua esperienza di militanza e la sua visione della città.
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Figure 3. The We-Z archive retrieval interface: a search for "trasformazione" returns 14 results across all three source types - oral history video interviews (OH), institutional archive documents (AS), and crowdsourced materials (CS) - interpretively connected through a shared conceptual metadata term. Credit: Manfredi Scanagatta / Progetto We-Z